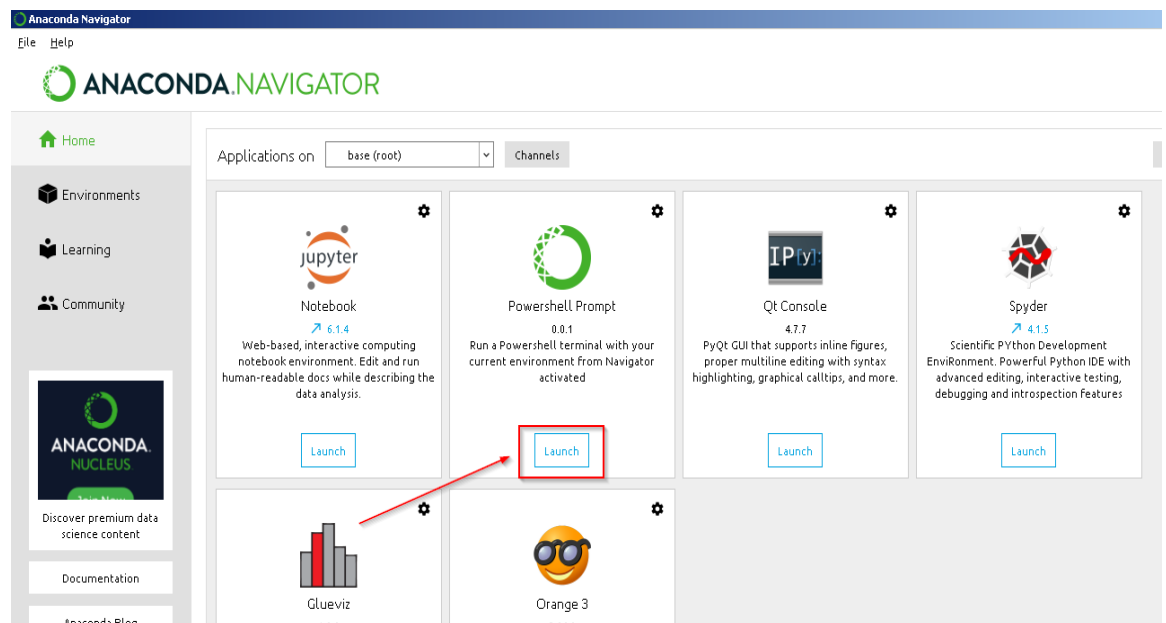
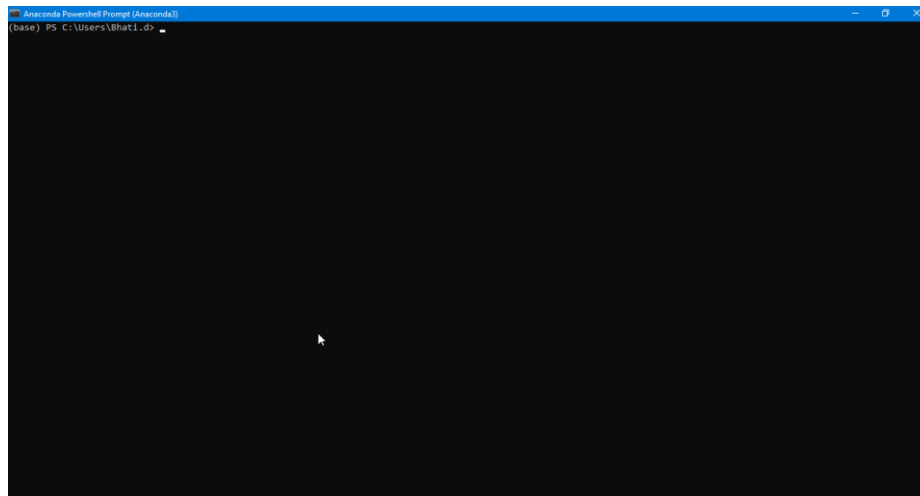


Assignment No 1

1. After successfully installation of Anaconda and OpenCV as stated in Lecture No 6 – Getting Started with Python and OpenCV, for quick reference have added the link below with steps and please follow the steps :
 - a. Download Anaconda (which includes Python for **Windows** in the given link and follow instruction for installation)
 - i. <https://docs.anaconda.com/anaconda/install/windows/>
 - ii. https://repo.anaconda.com/archive/Anaconda3-2020.02-Windows-x86_64.exe
→ **DOWNLOAD Link**
 - b. Download Anaconda (which includes Python for **MacOS** in the given link and follow instruction for installation)
 - i. <https://docs.anaconda.com/anaconda/install/mac-os/> → macOS graphical install
 - ii. https://repo.anaconda.com/archive/Anaconda3-2020.02-MacOSX-x86_64.pkg → **DOWNLOAD link**
 - c. [System Requirements](#)
 - d. OpenCV Installation steps :
 - i. Open Anaconda Navigator, something like this window would open up and Launch Powershell Prompt



- ii. After launching Powershell Prompt, will open up this type of window as shown below

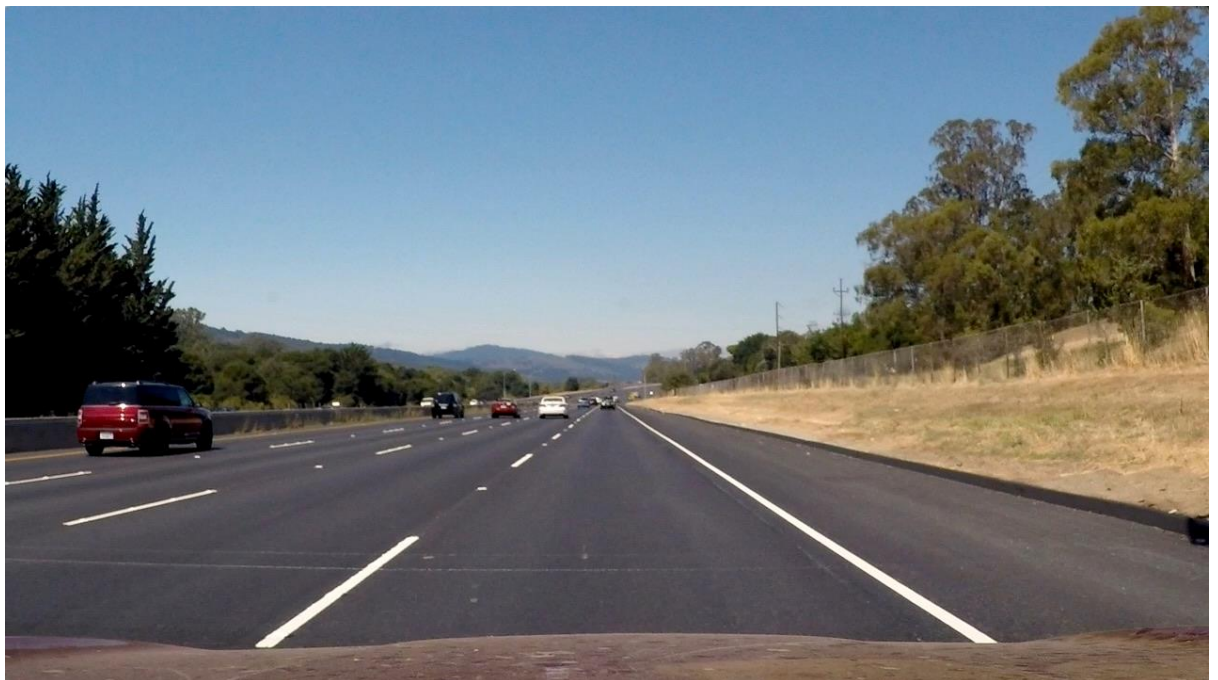


- iii. Write the below mentioned command in powershell prompt, as shown below

pip install opencv-python

```
(base) PS C:\Users\Bhati1.d> pip install opencv-python
Collecting opencv-python
  Downloading opencv_python-4.5.1.48-cp37-cp37m-win_amd64.whl (34.9 MB)
    | 34.9 MB 6.4 MB/s
Requirement already satisfied: numpy>=1.14.5 in c:\users\bhati1.d\anaconda3\lib\site-packages (from opencv-python) (1.18.1)
Installing collected packages: opencv-python
Successfully installed opencv-python-4.5.1.48
```

2. After Successfully creating the Software Environment, then please use the attached image Sample1.jpg below



1. Python Script with OpenCV library to perform below tasks –
 - I. Draw Red color Rectangle Boxes on Red Suv and White Car
 - II. Write label text upon Rectangular Boxes as “Red Suv” and “White Car” with Blue Color
 - III. Draw the straight Lane lines with Green color
2. Submit the Python Script along with Final output image with all the above tasks.

Note : In order to view the image in the Spyder Environment using matplotlib library, there would be color space conversion required to view **if the image is imported using OpenCV**, since in OpenCV color space is in BGR format and in matplotlib its in RGB format.

We need to convert the color space as shown below in sample script -

```
import cv2
import matplotlib.pyplot as plt

image_cv2 = cv2.imread('sample.jpg')
image = cv2.cvtColor(image_cv2, cv2.COLOR_BGR2RGB)
image_plt = plt.imshow(image)
```